

UNIT 4: MACHINE LEARNING – DECISION BOUNDARY & COST FUNCTION OPTIMIZATION

1. Decision Boundary

Definition:

A decision boundary is a line, curve, or surface that separates different classes in a dataset.

- In 2D $\rightarrow$ line/curve
- In 3D $\rightarrow$ plane
- In higher dimensions $\rightarrow$ hyperplane

Mathematical Form (Linear):

$$w_1x_1 + w_2x_2 + b = 0$$

Where:

- $w_1, w_2 \rightarrow$ weights
- $b \rightarrow$ bias

Logistic Regression Decision Boundary:

$$P(y=1) = 1 / (1 + e^{-(w^T x + b)})$$

Boundary occurs when:

$$w^T x + b = 0$$

Types of Decision Boundaries:

1. Linear Boundary
 - Straight line
 - Used in Logistic Regression, Linear SVM
2. Non-Linear Boundary
 - Curved
 - Used in Decision Trees, Neural Networks

Example:

x_1 x_2 Class

2 3 0

4 5 0

6 7 1

8 9 1

A line like:

$x_2 = x_1 + 1$ separates the classes.

Key Points:

- Separates classification regions
- Depends on model parameters
- Learned during training

2. Cost Function Optimization

Definition:

A cost function measures how incorrect the model predictions are.

Goal:

Minimize the cost function

$J(\theta) \rightarrow \text{minimum}$

Mean Squared Error (MSE):

$$J = (1/n) \sum (y_{\text{pred}} - y_{\text{actual}})^2$$

Example:

Actual: 10, 20

Predicted: 8, 22

$$\text{MSE} = [(10-8)^2 + (20-22)^2]/2 = 4$$

Log Loss (Binary Cross Entropy):

$$J = -1/n \sum [y \log(p) + (1-y) \log(1-p)]$$

Used in classification problems.

Gradient Descent

Formula:

$$\theta = \theta - \alpha (dJ/d\theta)$$

Where:

- α = learning rate
- $dJ/d\theta$ = gradient

Types of Gradient Descent:

1. Batch Gradient Descent
 - Uses full dataset
 - Accurate but slow
2. Stochastic Gradient Descent
 - Uses one data point
 - Fast but noisy
3. Mini-batch Gradient Descent
 - Balanced approach

Example:

$$J(w) = (w - 5)^2$$

Minimum occurs at: $w = 5$

Learning Rate (α)

- Small $\alpha \rightarrow$ slow learning
- Large $\alpha \rightarrow$ unstable

Challenges in Optimization

- Local minima
- Saddle points
- Overfitting

Regularization

$$J = \text{Loss} + \lambda \sum w^2$$

Prevents overfitting by penalizing large weights.

Connection Between Topics

- Cost function adjusts parameters
- Parameters define decision boundary

Therefore:

Minimizing cost $\rightarrow$ Better decision boundary

Summary

- Decision boundary separates classes
- Cost function measures prediction error
- Gradient descent minimizes cost
- Optimization improves model accuracy